

5(a)	amplitude of the carrier wave varies	M1
	in synchrony with the displacement of the (information) signal	A1
5(b)(i)	wavelength = $(3.0 \times 10^8) / (300 \times 10^3)$ = 1000 m	A1
5(b)(ii)	bandwidth = 16 kHz	A1
5(b)(iii)	frequency = 8 kHz	A1
5(c)	attenuation = $10 \lg (P_1 / P_2)$	C1
	$73 = 10 \lg (P_T / P_R)$	C1
	$73 = 10 \lg (P_T x^2 / 0.082 P_T)$ or $x^2 / 0.082 = 10^{7.3}$	
	$x = 1300$ m	A1